

Co-simulation: 1D to 3D coupling in an underground mine ventilation simulated model.

Nestor Rueda^{1*}

¹*Isaeng, The Netherlands - Colombia*

*Corresponding author: nestor.rueda@isaeng.com

In this paper, the author presents a procedure for a co-simulation model applied to a sub-level stoping mine. A two-way approach is presented which can be summarized as follows: a network model is implemented in the ventilation software Ventsim and coupled with a 3D computational fluid dynamics (CFD) model. Pressure and flows are calculated in the network model and used as inputs to the 3D model, once the CFD simulation is finished, the pressure and flow are sent back to the network model as inputs from the 3D model.

Introduction

Network ventilation modeling in underground mining became easier and more robust in the last years with the help of many commercial software options available in the market [1]. Recirculation, gasses, fog, fire, buoyancy and energy analysis are possible nowadays, even with modest computing and financial resources. Thanks to these software tools it is now possible to model bigger and more complex underground mine networks taking multiple scenarios and variables into account. Thanks to the network modeling approach, one of the advantages of the available tools is the short running time it takes to simulate even the most complex network. It usually takes only seconds for most common configurations, perhaps a couple of minutes in the case of transient thermodynamic analysis or fire analysis. However, as convenient as it might be, network modeling still imposes certain limitations in some important scenarios, as for example gas dilution in the development front or big voids in many of the different mining methods. These are the scenarios that require a detailed 3D simulation to capture the relevant details.

Nowadays, Computational Fluid Dynamics (CFD) is used to simulate 3D flows; it is a mature technology widely used in aeronautics, automotive and many other industries for upfront design, validation, and optimization. There are many available publications of the use of CFD in underground mining, but most of them are applied to academic or research problems. Industrial applications are not published often, and there is still a lack of confidence in or familiarity with the CFD method in the mining community. Some drawbacks of the CFD approach, at least when compared with the most familiar network modeling analysis is the time required to set up and simulate a single case, the cost of the software and hardware and the lack of knowledge among the practicing engineers.

This paper presents the methodology of a balanced approach to decrease the gap between the network and the CFD models known as co-simulation. This approach has been successfully used in some vehicular tunnels [2] as well as in coal gob analysis [3]. In both cases, this approach was called multi-scale analysis. Among the many benefits of co-simulation are, reduction of time many orders of magnitude when compared with a pure CFD approach as well as a reduction of cost since the author uses a well-known Open Source CFD code.

Methodology

Network model

The network model is implemented in Ventsim, a well-known underground mine ventilation software. The procedure to model any network can be summarized as follows: 1) Creation (using ventilation software) or import of a CAD model representing the centerlines of the tunnels, 2) Definition of the geometric properties of the different tunnels, 3) Connections to the surface and definition of closed ends, 4) Initial proof of concept of the mine, this step usually ensures that all the airways are correctly interconnected. This can be easily done with a natural ventilation simulation pass, 5) Definition of operating parameters, conditions and equipment such as fans, ducts, doors, blockades, diesel equipment, etc., 6) Test-run and assessment of the model, 7) Validation against field measurements, 8) Final adjustments and optimization of the model. Independent of the mining method, in a network model all the tunnels or voids in the mine, are treated as 1D airways. This has proven to give good results even for big voids like those present in sublevel stoping. However, one of the cases where the network approach falls short is with the modeling of gas dilution after a blast event, especially because of the accumulation of gasses in the

recirculating areas of the cave. 1D networks can incorporate a dilution factor, but the determination of the factor is mostly by trial and error. The impact of this situation on production is very high because of the delayed or even lost shifts until the gas levels are safe.

CFD model

Similarly, to 1D network models, the principle behind CFD relies on the discretization or division of the physical domain into multiple and smaller elements or volumes to numerically solve the equations of Navier-Stokes. These are a set of non-linear equations describing the flow behavior in time and space. The most common numerical approach to solve these equations is known as the finite volume method or FVM. The model for this case is simulated in CAELUS [4], a version of a popular Open Source CFD code called OpenFOAM. A common CFD workflow can be summarized as follows: 1) Import or creation (using a pre-processing software) of a CAD model representing the domain, 2) Discretization or meshing of the model, 3) Definition of operating parameters and conditions, 4) Simulation set-up, 4) Test-run and assessment of the model, 5) Validation and verification of the model. The configuration of a CFD model is far more complicated and time-consuming than that of a 1D network model. It also requires more resources as well as time to be solved with a decent accuracy resulting in a large difference in the solution times of a 1D network model compared with a 3D CFD model.

The author has applied CFD to model the scenario described in the previous section to try to account for the gas dilution in the cave after a blasting procedure. These procedures usually involve the simulation of the network model first and then using the resultant flows and pressures entering the cave as inputs to the 3D CFD model. Although totally viable, this approach is done manually, it is, time-consuming and prone to errors. In this case, co-simulation offers, a very good balance between performance and accuracy.

Co-simulation

Co-simulation is a technique to connect different spatial domains or sub-systems. It also refers to the coupling and transfer of data between codes performing analysis in different physics domains [5]. Applications span a wide range of industries from aeronautics to health sciences. In tunnel ventilation, there are interesting applications of this technique in vehicular tunnels [2], [6]. However, the only one reference of co-simulation the author found in underground mining ventilation is reported in coal gob characterization by Wedding [3]. These important works refer to co-simulation as multi-scale simulations. However, in the field of computational mechanics, multi-scale is better explained as a procedure in which the continuum assumption is no longer enough to model the different physical/spatial domains involved [7]. Multi-scale analysis scope goes from the micro-scale (atomic or molecular level) to the macro-scale (astronomical level). Co-simulation, as shown in detail in the following sections, is an effective alternative that takes advantage of the 1D network approach and detailed 3D CFD modeling.

Case set up

Case description

Let us take a case of a sub-level stoping limestone mine. The mine is in the department of Antioquia in Colombia. It has four production levels, but only level 3 is in exploitation. The stopes or caves are on average 100 m long, 30 meters wide and 45 meters high [Figure 1]. The mine can use as much as 2000 kg of ANFO per blast in the stopes and use up to 350 kg of ANFO per blast in the advancing fronts. After each blast, the gasses contaminate the atmosphere in the mine. Because of the many open caves, the gasses generated during the blast tend to accumulate and take longer than desired to be extracted from the mine. Sometimes it can cause a loss of a complete shift.

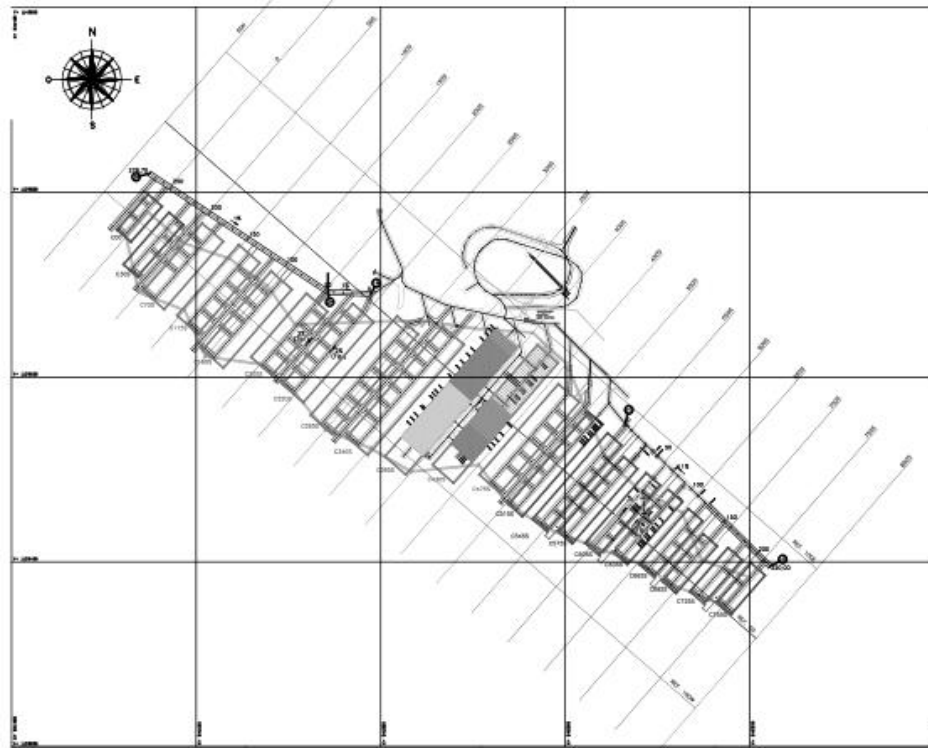


Figure 1. Schematic of the production level.

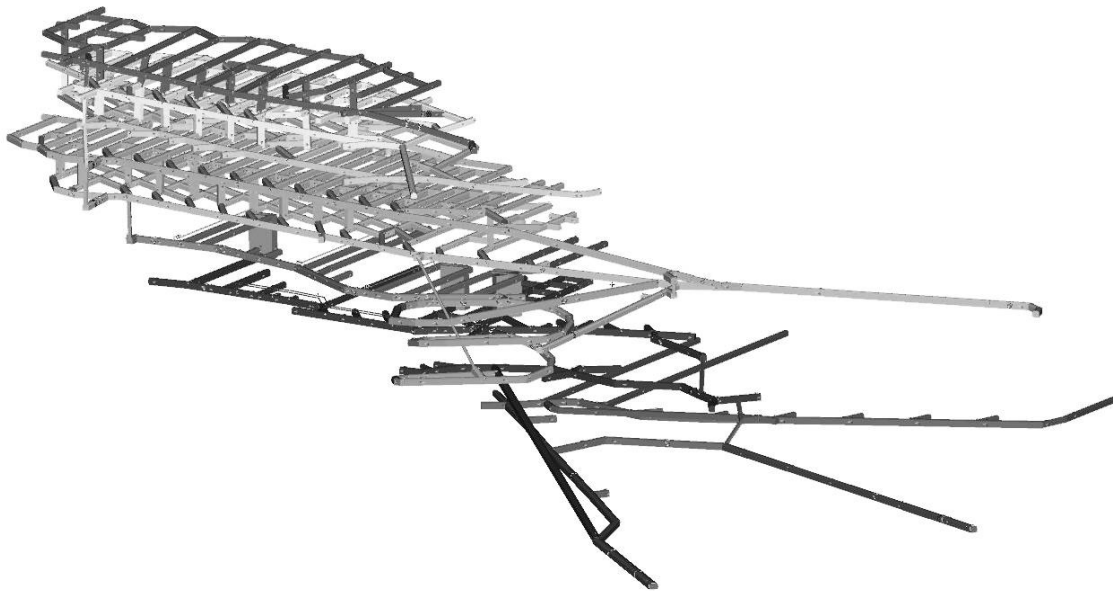


Figure 2. Ventsim model of the mine.

Computational set up

The network model of the mine is already available in Ventsim [Figure 2]. The stopes are modeled with the approximate height and width of the actual ones [Figure 3]. All the operating conditions and equipment are also incorporated in the model, and the simulation is performed as usual.

For this case, the interaction boundaries are defined at the access tunnels of the stope. These boundaries need to be carefully chosen, defining them too close to the stope will affect the results of the simulation, too far will unnecessarily increase the simulation time in the CFD model. There is not a standard procedure to choose this distance, so this process depends on the problem, engineering judgment and error estimation.

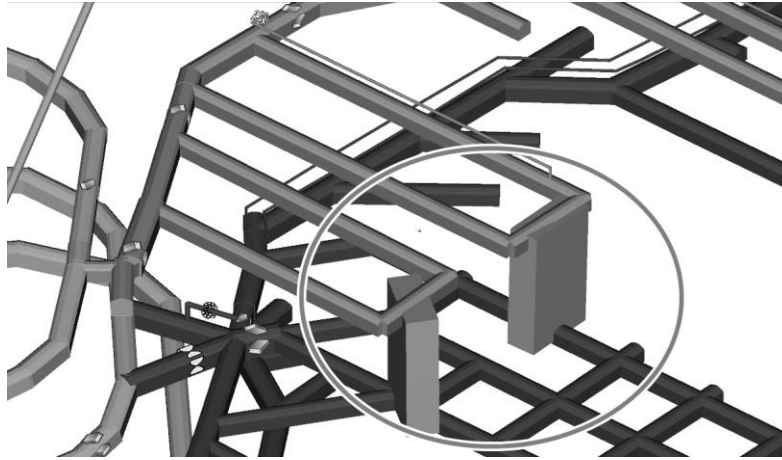


Figure 3 Simplification of the network model.

The model of a cave is modeled in the CAD program ANSYS SpaceClaim and the meshing is created in ANSYS Workbench and exported in fluent text format to be read by the CFD software Caelus [Figure 4].

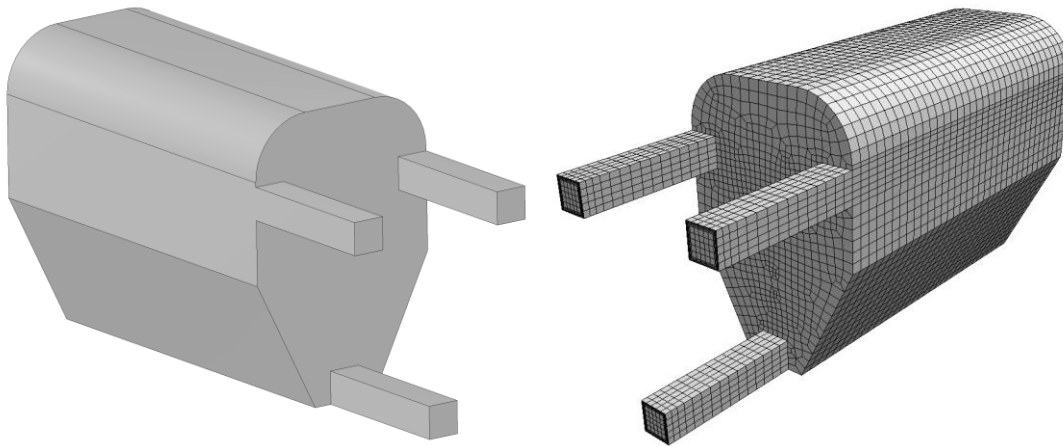


Figure 4. 3D model and mesh of the chamber for CFD simulation.

Boundary conditions for the CFD models are:

- Inlets: mass flow (Calculated from Ventsim [Figure 5])
- Outlet: pressure outlet
- Walls: zero velocity

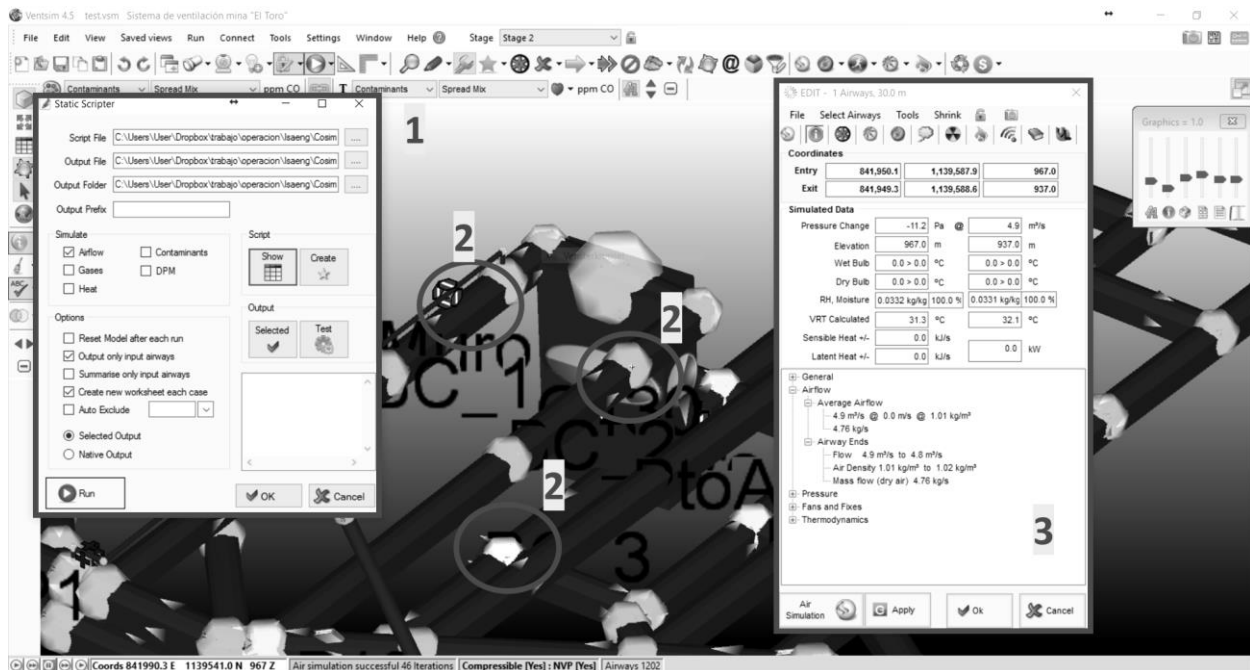


Figure 5. Description of the different boundaries and commands used to exchange information from Ventsim to the CFD package: 1. Static scripiter window, 2. Inlet and Outlet boundaries, 3. General information window with the mass flow and pressure values.

The co-simulation procedure is executed with the help of a script created in the Python programming language. This script is implemented to communicate both models sequentially during the simulation process. The co-simulation routine can be described as follows:

1. Run Ventsim;
2. Write results from network model (Ventsim output file);
3. Update CFD boundary conditions (CFD input file);
4. Run CFD;
5. Write results from CFD results file (CFD output file);
6. Update Ventsim boundary conditions (Ventsim input file);
7. Go back to step 1 until convergence.

The results at the interaction boundaries are marked for storing in output files during steps 2 and 5. These files will be read later by each code to sequentially update its own boundary conditions during steps 3 and 6. This procedure needs to be done multiple times until the exchanging values in the network and CFD model converged to a defined value.

In principle, any variable can be exchanged, there are, however, some practical limitations. It is not recommended to use the calculated flow from the CFD model as an input in the network model since it can affect the stability of the simulation. Instead, it is more robust to use the calculated pressure from the CFD model, which is updated in the network as a flow resistance at the defined boundaries. This resistance will update the flow which will be stored as input for the CFD model. In the case presented here, resistances were defined at the two upper tunnels entering the stope in the Ventsim model.

For CFD analysis, the most robust combination of boundary conditions is usually achieved by defining flow at the inlet boundary conditions, while defining pressures at the outlet boundary conditions. In this sense, the calculated flows from Ventsim are used as inputs in the CFD model. These defined flows, as well as the defined pressures at the outlet, will allow calculating the pressure at the inlets which will be used as resistances in the Ventsim model.

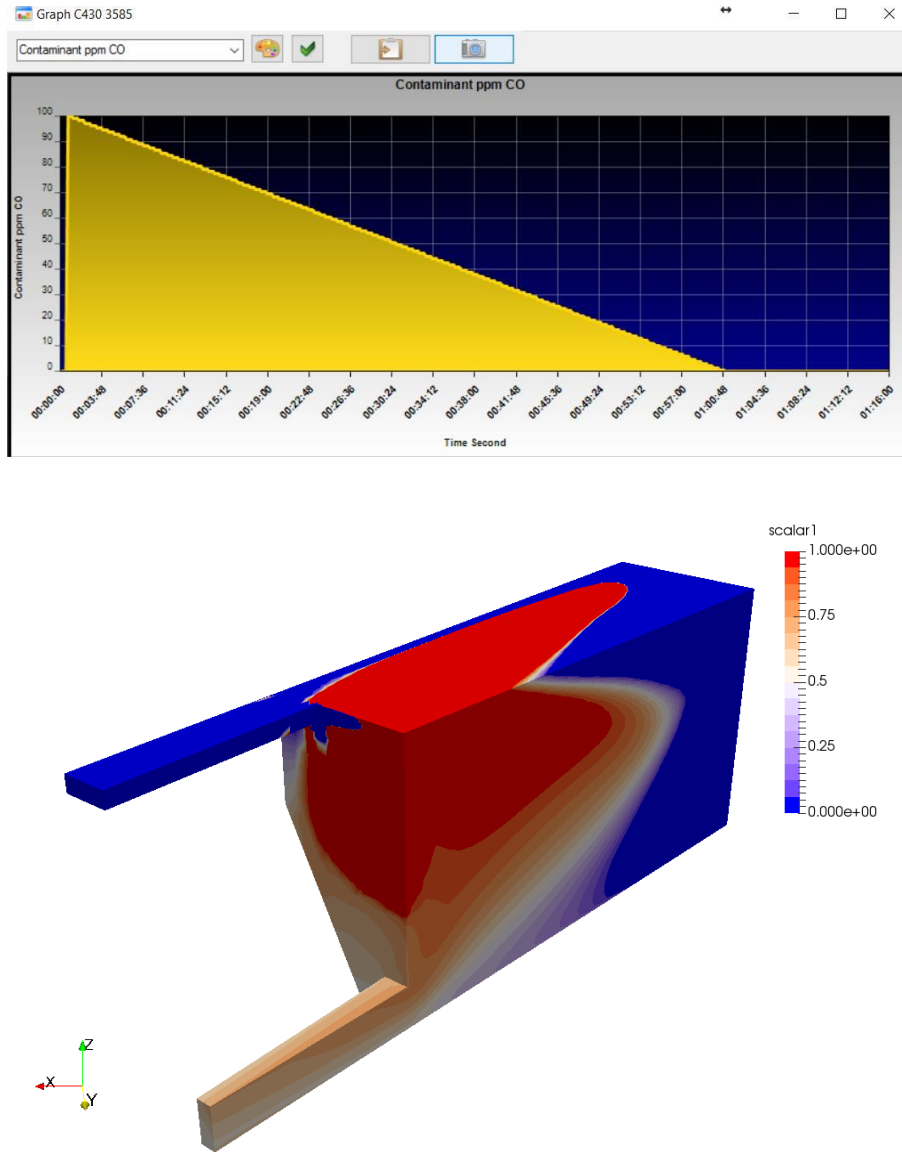


Figure 6. Contaminant concentration after 1 hour for the Ventsim model and for the 3D CFD model.

Figure 6 present the comparative results of concentration in the stope after one hour. In this case the 3D CFD model shows that after one hour of ventilation there is still a high concentration of contaminant in the stope. It is important to mention that the Ventsim manual warns that dynamic contamination should be used as a guide for estimation. This is because of the simplified homogeneous distribution of the 1D network.

Discussion

An automated co-simulation process was possible with minimum programming effort. More complex simulation scenarios can be evaluated depending on the capabilities of both simulation codes.

The simulation time for the CFD is less than it would take for a full 3D CFD model, due to using co-simulation, there is still space for improvement, especially through new and more advanced numerical methods. One of the more robust methods is known as Reduced Order Model (ROM). In a study by Vervecken et al. [8] they demonstrate how ROM brings down the simulation time in a CFD simulation by several orders of magnitude. The author is taking further efforts to implement this technique in the co-simulation strategy presented here.

Simplification might be at the expense of accuracy. One of the most critical aspects of the simulation models is the wall roughness. It is necessary to account for a correct pressure loss in the ventilation network. With the help of new scanning technologies, it is possible now to implement an even more detailed 3D model of the mine. This application should be further discussed and studied.

There are no available validations of CFD models applied to underground mine ventilation. Validation of computational models is vital to promote the safe use of CFD in the mining environment. Associations such as NAFEMS, ASME or AIAA have made a great effort to quantify the quality of numerical simulations. It is important to promote validation and verification initiatives within the underground mine ventilation community, including the academia and the industry.

The author considers it important that the ventilation community refers to 1D to 2D or 3D coupling as co-simulation instead of multi-scale. The term co-simulation is more consistent with existing definitions and can help differentiate from the already complex field of multi-scale physics.

Conclusions

The author summarizes a co-simulation strategy applied to underground mine ventilations problems which offer a balance between accuracy and computational resources. Minimum programming skills are required to implement this strategy since it takes advantage of already available resources like Ventsim, Caelus, and Python. The location of the boundaries, as well as the definition of the parameters to exchange, are the most important factors to consider for the co-simulation strategy to work properly. The proper validation of the models, as well as the incorporation of advanced numerical methods, will allow for further optimization of the performance and accuracy of the co-simulation process.

References

- [1] W. Keith, P. Brian, and S. J. Daniel, "International Journal of Mining Science and Technology The practice of mine ventilation engineering," *Int. J. Min. Sci. Technol.*, vol. 25, no. 2, pp. 165–169, 2015.
- [2] F. Colella, G. Rein, R. Carvel, P. Reszka, and J. L. Torero, "Analysis of the ventilation systems in the Dartford tunnels using a multi-scale modelling approach," *Tunn. Undergr. Sp. Technol. Inc. Trenchless Technol. Res.*, vol. 25, no. 4, pp. 423–432, 2010.
- [3] W. C. Wedding, "Multiscale modeling of the mine ventilation system and flow through the gob," University of Kentucky, 2014.
- [4] Appliedccm, "CAELUS." 2016.
- [5] S. A. Sicklinger, "Stabilized co-simulation of coupled problems including fields and signals," Technische Universität München, 2014.
- [6] F. Colella, G. Rein, R. Borchellini, and J. L. Torero, "A Novel Multiscale Methodology for Simulating Tunnel Ventilation Flows During Fires," *Fire Technol.*, vol. 47, no. 1, pp. 221–253, 2011.
- [7] L. Wright, "Multi-Scale Modelling – Bridging Different Scales." NAFEMS, 2016.
- [8] L. Vervecken, J. Camps, and J. Meyers, "Stable reduced-order models for pollutant dispersion in the built environment," *Build. Environ.*, vol. 92, pp. 360–367, 2015.